Def. A function $f: G \to \mathbb{C}$ defined on an open set $G \subseteq \mathbb{C}$ is called Meromorphic if for some finite number points $z_1, \cdots, z_n \in G$,

f is Holomorphic on $G \setminus \{z_1, \cdots, z_n\}$ and f has pole at $z \in \{z_1, \cdots, z_n\}$.

Thm. Every Meromorphic function on Extended Complex plane is Rational function.

Proof Let f be a Meromorphic function on Extended Complex plane.

That is, f is Meromorphic on $\mathbb{C}$ and the function $z \mapsto f(1/z)$ has a pole or removable singular at $z = 0$.